# HARDWARE II

Week 2

# MORE ON CACHE

- In a CPU cache, frequently used code or data is loaded in chunks (quite a large amount of something) from the slower RAM into the cache.
- The CPU accesses the cache memory at its own, faster speed.
- This means the CPU isn't slowed down by having to wait for data from the RAM.

# CACHE MISS

when the data requested for processing by a component or application is not found in the cache memory

# VIRTUAL MEMORY USE CASE

When a computer is running the operating system and several applications at the same time, the RAM often becomes full. Instead of closing some programs, the 'memory manager' program of the operating system will use 'pretend' or virtual memory to store some of the data, usually on the hard disk drive.

# DISADVANTAGES OF VIRTUAL MEMORY

- The read/write speed of a hard drive is much slower than RAM, and the technology of a hard drive is not made for accessing small pieces of data at a time.
- If the system has to rely too heavily on virtual memory, there will be a significant drop in performance.
- Often the operating system has to constantly swap information back and forth between RAM and the hard disk drive, which operates all of the time. You can hear the disk drive operating continuously. This is called 'disk thrashing' and it slows down the execution of the programs.

# ACTIVITY

1. What is cache memory used for?
2. hat is the difference between RAM and ROM? Suggest a use for ROM.
3. Why would increasing the amount of RAM improve the performance of a computer?

# FETCH-DECODE-EXECUTE IN DETAIL

# INSIDE THE CPU.

- Arithmetic/logic unit (ALU): the part of the CPU that performs calculations and logic operations.
- Register: a storage location inside the CPU used to hold an instruction, an address or other single item of data.
- Control unit: the part of the CPU that organises the actions of the other parts of the CPU

# COMMON REGISTERS

- Accumulator: this holds the results of calculations performed by the ALU. All input and output from the CPU pass through the accumulator.
- Program counter: this holds the memory address of the next instruction to be fetched.
- Current instruction: this holds the instruction currently being executed.

# MORE ON CPU

- Clock: an electronic device inside a CPU that 'ticks' at regular intervals and is used to synchronise the actions of the other parts of the CPU
- gigahertz (GHz) a measure of frequency equivalent to 1000 million cycles per second
- Bus width: the number of wires that make up a bus. This determines the range of binary numbers that can be communicated

# FETCH-DECODE-EXECUTE CYCLE

- The CPU control unit **fetches** the next instruction by placing its memory address on the address bus and signaling the memory to read. The data is then copied to the data bus and into a special register. During the decode phase, the control unit analyzes the register's contents and instructs other CPU components accordingly. Finally, the CPU executes the instruction.